
TriMet Data Pipeline

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Team: <Work_in_Progress>

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System Description

BreadCrumbs

GPS points every ~5 seconds



TriMet Buses

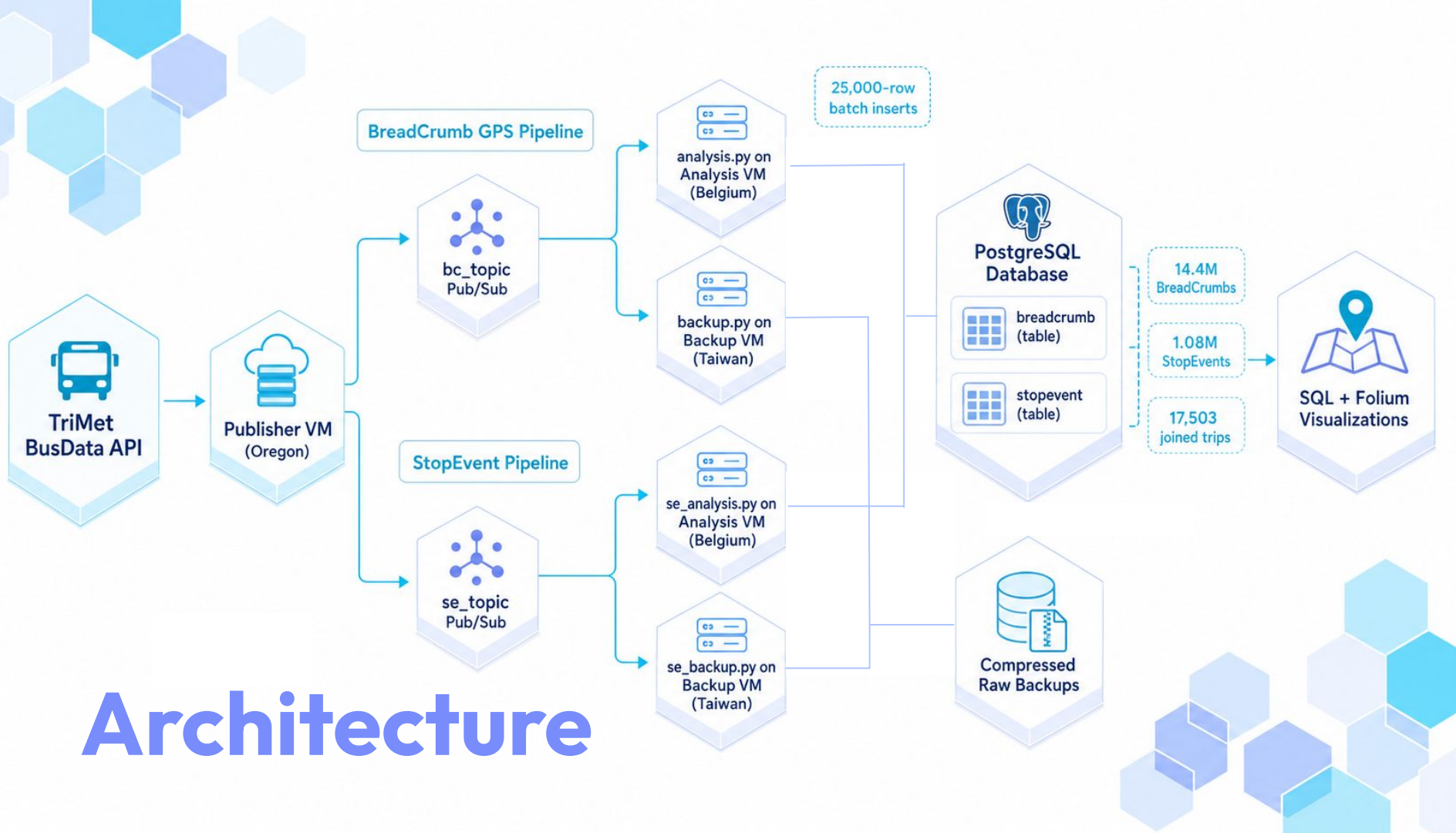
StopEvents

stop arrivals +
passenger activity

TriMet is the regional transit organization for the metropolitan area of Portland, Oregon. It operates a fleet of 600+ buses servicing over 80 routes in the Portland metro area.

BreadCrumbs: GPS telemetry records that capture a bus's position, timestamp, trip ID, vehicle ID, and odometer reading at roughly 5-second intervals.

StopEvents: Per-stop transit event records that capture when a bus arrives/leaves a scheduled stop, along with route, trip, stop location, dwell time, and passenger activity such as boardings and alightings.



Data Statistics

Metric	BreadCrumbs	StopEvents
Total Records	14,369,896	1,079,671
Table Size	1050 MB	278 MB
Service days	20	14
Weekday average	820,727 /day	81,944 /day
Weekend average	411,799 /day	65,058 /day

Visualizations



Challenges and Lessons

1. Hardware Constraints
 - Memory issues mean I had to use buffers, tune hyperparameters like flow control.
 - 25,000 buffer and staging table for performance.
2. Data permanence
 - Should have focused on solved backup VM early and always left running to ensure having lots of data.
 - Sometimes GCP fails, and should have set up an alert so I'd know.

General comment: I liked the project, although ending felt a little incomplete. I think perhaps an MCP server taking natural language to execute SQL queries and producing visualizations about the database would be interesting approach.

THANK YOU

Trimet for the awesome data

Bruce for the class

And anyone watching :)

